



Cellular therapies for rheumatic disease

Fotios Koumpouras^a and Roberto Caricchio^b

Purpose of review

Cellular therapies, particularly chimeric antigen receptor (CAR)-modified lymphocytes, have progressed from experimental oncology into serious consideration for selected autoimmune rheumatic diseases. Autologous and emerging allogeneic CAR-T platforms now offer the possibility of deep, drug-free remission in systemic lupus erythematosus (SLE), idiopathic inflammatory myopathies (IIM), systemic sclerosis (SSc), and related conditions for which conventional therapies remain inadequate. This timely article reviews the rationale, current clinical experience, safety profile, and future directions of cell therapies in rheumatology with a focus on efficacy, safety, and “immune reset” in autoimmune rheumatic disease. The review is particularly pertinent as multiple parallel cell-based platforms (autologous and allogeneic CAR T cells, transient RNA CARs, CAR-NK, and T cell engagers) are entering the rheumatology space faster than practice guidelines or trial frameworks can fully adjust.

Recent findings

Recent studies show that CD19-directed CAR T cells can induce deep B-cell depletion with high rates of drug-free remission in refractory SLE and promising responses in SSc and IIM, accompanied by distinctive toxicity patterns such as mostly low-grade CRS, rare ICANS, and the newly described organ-specific LICATS. Parallel work demonstrates mechanistic “immune reset” (including type I IFN pathway suppression and naïve-skewed B-cell repopulation), expansion of indications to neurologic autoimmunity, emergence of off-the-shelf platforms (allogeneic CARs, $\gamma\delta$ -CAR, CAR-NK, RNA CARs), and early human experience with CD19- and BCMA-directed T cell engagers in rheumatic disease.

Summary

Collectively, these findings position cellular therapies as powerful, potentially transformative options for highly selected patients with severe, refractory lupus and autoimmune rheumatic disease, but also underscore the need for disciplined trial design, long-term safety surveillance, and strategies to ensure equitable access.

Keywords

autoimmune rheumatic disease, bi-specific t-cell engagers, chimeric antigen receptor T, cellular therapies, systemic lupus erythematosus

FOUNDATIONS OF CELLULAR IMMUNOTHERAPY AND CHIMERIC ANTIGEN RECEPTOR DESIGN

The modern era of cellular immunotherapy began with studies showing that ex vivo-expanded lymphocytes could mediate regression of metastatic tumors when reinfused into patients [1[■],2,3]. These early efforts included systemic transfer of interleukin-2-expanded lymphoid cells and the use of tumor-infiltrating lymphocytes combined with high-dose cytokine therapy [1[■],2,3]. Subsequent work demonstrated that retroviral gene transfer could endow autologous T cells with tumor-reactive receptors, inaugurating the concept of genetically engineered cell therapy [4,5[■],6].

Chimeric antigen receptor (CAR)-T technology extends this framework by combining an extracellular single-chain variable fragment that recognizes a defined antigen with intracellular CD3 ζ and one or more co-stimulatory domains, typically CD28 or

4-1BB. These design elements determine activation thresholds, proliferation, persistence, and cytokine secretion. CD28-based CARs often show brisk expansion but shorter persistence, whereas 4-1BB

^aSection of Rheumatology Allergy and Immunology, Department of Medicine, Yale School of Medicine, New Haven, Connecticut and ^bDivision of Rheumatology, Department of Internal Medicine, University of Massachusetts Chan Medical School, Worcester, Massachusetts, USA

Correspondence to Fotios Koumpouras, MD, Associate Professor of Medicine, Department of Medicine, Section of Rheumatology, Allergy & Immunology, Yale School of Medicine, The Anlyan Center S525b, 300 Cedar Street, New Haven, CT 06520, USA. Tel: +1 203 737 2771; e-mail: fotios.koumpouras@yale.edu

Curr Opin Rheumatol 2026, 38:155–162

DOI:10.1097/BOR.0000000000001159

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KEY POINTS

- **Advancements in cellular therapies:**
Chimeric antigen receptor (CAR) T-cell therapies have transitioned from oncology to autoimmune rheumatic diseases. This advancement represents a significant shift as autologous and emerging allogeneic CAR T platforms show potential in inducing deep, drug-free remissions in diseases such as systemic lupus erythematosus (SLE), idiopathic inflammatory myopathies (IIM), and systemic sclerosis (SSc). Main targets include CD19, BCMA, CD20.
- **Challenges in managing adverse effects:**
Patients undergoing CAR T treatment can experience severe adverse effects such as cytokine release syndrome (CRS), immune effector cell-associated neurotoxicity syndrome (ICANS). The occurrence of these conditions, along with local immune effector cell-associated toxicity syndrome (LICATs), has made the management of these therapies more complex given the need for comprehensive management strategies to mitigate these risks.
- **Concept of immune reset:**
The concept of immune reset is crucial in understanding the long-term efficacy of CAR T-cell therapies. Immune reset refers to the idea of reprogramming the immune system to eliminate pathological autoimmunity while restoring normal immune function. This needs to be better defined but thus far is demonstrated by presence of CD19 B cell depletion and type-1 interferon pathway suppression, followed by repopulation with a naïve B cell phenotype in association to major clinical response.
- **Future directions and optimizations:**
Future strategies include innovations such as dual-targeting CARs, allogeneic products and bi-specific T-cell engagers to maximize access and therapeutic benefits while minimizing adverse effects.
A significant challenge in the development of these therapies is the overabundance of concurrent clinical trials, which dilutes patient populations and stretches resources thin. To address this, innovative trial designs matching individuals with the most appropriate trials based on their unique genetic, phenotypic, and disease characteristics should be considered.

constructs expand more slowly but persist longer in vivo. Most autoimmune CAR-T programs have therefore adopted CD19–4-1BB architectures to balance robust B-cell depletion with manageable toxicity [7,8,9[■],27,28,29[■],30[■],36[■],37–40,41[■],42–44,46].

PERSISTENT UNMET NEED IN SYSTEMIC LUPUS ERYTHEMATOSUS AND RELATED DISEASES

Despite important advances, treatment of systemic lupus erythematosus (SLE) and lupus nephritis

remains unsatisfactory for many patients. Conventional immunosuppressive regimens and newer targeted therapies such as interferon receptor blockade, B-cell activating factor (BAFF) inhibition, and calcineurin inhibitors have improved outcomes but often fail to induce durable, complete renal remissions, and cumulative glucocorticoid exposure continues to drive organ damage and comorbidity [9[■],10–17]. Attempts to taper or discontinue immunosuppression in proliferative lupus nephritis are frequently limited by relapse risk, reflecting the difficulty of truly resetting the autoimmune response [15–19].

B-cell-targeted biologics highlight both the promise and limits of antibody-mediated approaches. Phase II/III rituximab trials in SLE did not meet primary endpoints despite clear serologic and peripheral B-cell effects in some subgroups. More potent type II anti-CD20 antibodies such as obinutuzumab achieve deeper and more sustained B-cell depletion and have shown positive results in proliferative lupus nephritis, yet complete, durable drug-free responses remain uncommon. These observations suggest that broader and more efficient targeting of the B-lineage compartment, including tissue-resident cells and plasmablasts, may be necessary for true immune “reset” in some patients, which provides a strong rationale for CD19-directed cell therapy [7,8,9[■],20[■],21,22,23[■],27,28,29[■],30[■],36[■],37–40,41[■],42–44,47[■],48,49[■],50].

Notably, a paradoxical unmet need is occurring in the context of a rapidly expanding development pipeline. As illustrated in Fig. 1, more than 30 interventional trials currently explore cell based therapeutic strategies in lupus nephritis and related lupus indications, reflecting substantial momentum and investment in immune reprogramming approaches alongside more established pharmacologic modalities.

WHY CD19-DIRECTED CHIMERIC ANTIGEN RECEPTOR-T IS COMPELLING IN AUTOIMMUNITY

CD19 is expressed on a wide spectrum of B-lineage cells from early developmental stages through naïve and memory B cells, plasmablasts, and a subset of short-lived plasma cells, whereas CD20 is absent from pro-B cells and many antibody-secreting cells [7,20[■],21,22,23[■]]. CD19 CAR-T cells can therefore deplete a broader swath of autoreactive B-cell and plasmablast pools, including populations that may be relatively inaccessible to monoclonal antibodies because of tissue localization or microenvironmental factors [7,8,9[■],20[■],21,22,23[■],27,28,29[■],30[■],37–40,41[■],42–44].

Preclinical lupus models support this concept: CD19–CAR-T therapy in murine systems can prolong

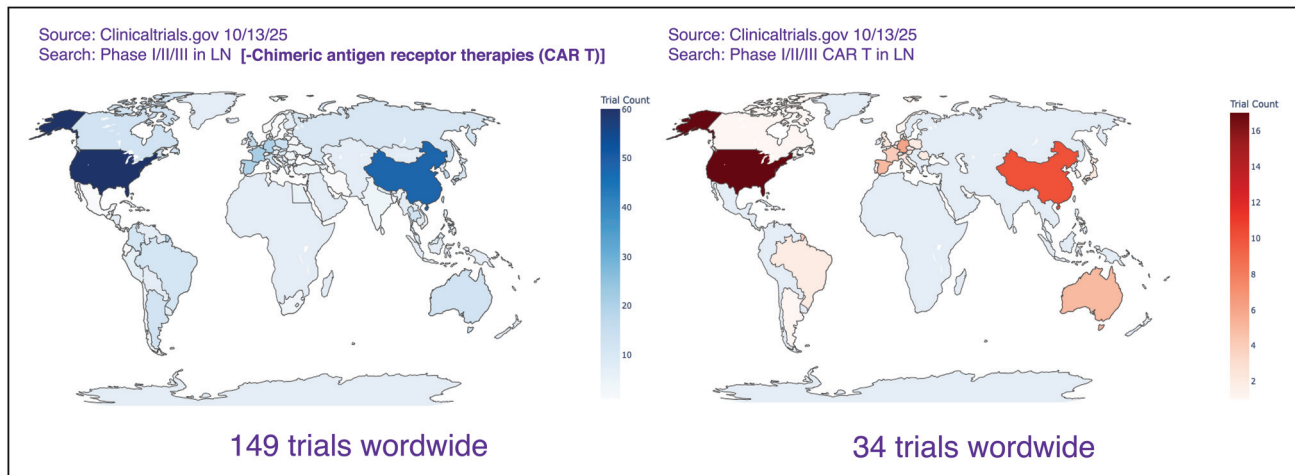


FIGURE 1. Ongoing interventional clinical trials in lupus nephritis by therapeutic mechanism. Summary of active interventional clinical trials in lupus nephritis as of 10/13/25, categorized by primary mechanism of action. Trials were identified through a structured search of public clinical trial registries via ClinicalTrials.gov using disease specific terms related to lupus nephritis and systemic lupus erythematosus, and were manually curated to exclude observational studies, supportive care interventions, and noninterventional designs. Therapeutic classes include B cell directed therapies, plasma cell targeting approaches, cytokine and interferon pathway inhibitors, complement inhibitors, intracellular signaling modulators, and cellular therapies. Trial counts reflect active registered studies at the time of analysis and are intended to describe the current mechanistic distribution of the development pipeline rather than comparative efficacy.

survival, prevent glomerulonephritis, and maintain long-term functional activity when B-cell depletion is sustained. In contrast, using the Fra2 TG murine model of systemic sclerosis, deep B cell depletion did not show efficacy but rather disease exacerbation [57].

In early human experience, leukapheresis and successful CAR-T manufacturing have been feasible even in patients maintained on immunosuppressive regimens, and ex vivo cytotoxicity of SLE-derived CAR-T products appear comparable to that of healthy donor-derived cells. These data support the practical viability of CAR-T approaches in heavily pre-treated autoimmune populations [27,28,29[■], 30[■],36[■],37–40,41[■],42–44].

CLINICAL EXPERIENCE WITH CD19 chimeric antigen receptor-T IN SLE, IIM, AND SSc

The first landmark report of CD19 CAR-T in autoimmunity was a single young adult with severe refractory SLE and active lupus nephritis who achieved rapid B-cell depletion and complete clinical and serologic remission after CD19–4-1BB CAR-T infusion, remaining disease-free off immunosuppression. This was followed by a five-patient series of refractory SLE with nephritis, in which all patients reached DORIS-defined drug-free remission within three months, maintained normal complement and

minimal or no proteinuria, and showed repopulation of the B-cell compartment predominantly by naïve cells.

Subsequently, the Erlangen group reported on 24 patients (17 women, 7 men) with a median (IQR) age of 39 years enrolled in a phase I/II basket study. 10 patients had SLE, 9 SSc and 5 IIM with a median number of 4 previous immunosuppressive treatments. 18/24 developed mild CRS (17 grade 1; 1 grade 2). No higher-grade CRS and no ICANS was observed. No clinically relevant neutropenia or thrombopenia >4 weeks were recorded. One grade 3 organ toxicity due to renal thrombotic microangiopathy in conjunction with CMV infection was reported. LICATs was reported in 88% of patients with grade 1/2 in the vast majority of patients. B-cells were depleted in all patients. Six-month efficacy data were available in 19/24 patients at data cut-off of May 12, 2025: All 7 SLE patients were in DORIS remission, all 8 SSc showed no disease progression and all 4 IIM patients met ACR/EULAR moderate or major response. In addition, clinical responses were also observed in the residual 5 patients, in whom follow-up time was still <6 months. All 24 patients discontinued immunosuppressive medications.

Aggregated analyses and 2024–2025 reviews of autoimmune rheumatic disease CAR-T cohorts now estimate that roughly 80–85% of SLE patients in early-phase trials achieve complete clinical responses, with the large majority in drug-free remission at six months

and many maintaining remission beyond one year [37–40,41[■],42–44]. Differentiating damage from disease activity, such as ongoing proteinuria in lupus nephritis related to damage, might make full clinical responses less apparent in these early studies. Smaller but consistent series in SSc and IIM report similar qualitative patterns of deep disease control after a single infusion, although longer follow-up and multi-center data are needed [27,28,29[■],30[■],36[■],37–40,47[■],48,49[■]].

TOXICITY AND SAFETY CONSIDERATIONS

Cytokine release syndrome (CRS) and immune-effector-cell-associated neurotoxicity syndrome (ICANS) are the key toxicities associated with CAR-T therapy (Table 1) [31–33,49[■]]. In hematologic malignancies, higher tumor burden correlates with more severe CRS and ICANS, often requiring intensive care and aggressive immunosuppression [24[■],25,26[■],31–33]. In contrast, early autoimmune cohorts have experienced predominantly grade 0–1 CRS, with occasional grade 2 events, and rare, usually low-grade ICANS manifesting as transient neurocognitive symptoms [29[■],30[■],31,36[■],37–40,41[■],42–44,49[■]].

Hematologic toxicities such as prolonged cytopenias appear less frequent and less severe in autoimmune disease than in oncology, with most patients recovering counts within weeks and no consistent signal for long-lasting pancytopenia to date [29[■],30[■],31,36[■],37–40,41[■],42–44]. Long-term hypogammaglobulinemia and infection risk remain critical concerns; however, early SLE CAR-T studies report relative preservation of pre-existing protective antibody titers to standard vaccines and the ability to boost responses upon re-vaccination, consistent

with sparing of CD19-negative long-lived plasma cells [29[■],30[■],36[■],37–40,41[■],42–44,49[■]].

Managing disease activity during immunosuppressive washout remains a practical challenge in CAR T cell therapy for SLE. We recently reported that brief pulse corticosteroids can control severe lupus flare during washout without compromising CAR T cell expansion, B cell depletion, or the achievement of durable, drug-free remission, supporting their cautious use as a bridging strategy in selected patients [58].

A recently described local immune effector cell-associated toxicity syndrome (LICATS) has been observed as a frequent, usually mild and self-limited, organ-specific toxicity occurring in 77% of 39 patients with severe SLE, systemic sclerosis, or idiopathic inflammatory myopathy treated with CD19 CAR T cells, manifesting only in organs previously affected by the underlying autoimmune disease, with 54 total events that most often involved skin (35%), kidneys (22%), and musculoskeletal system (19%), typically arising around 10 days after infusion during B-cell aplasia and resolving over about 11 days, usually without intensive therapy and often with only brief glucocorticoid courses [49[■],51[■]]. LICATS appear to be temporally and clinically distinct from early-onset systemic CRS, shows no serologic features of autoimmune flare or histologic evidence of active autoimmunity in limited biopsies, and is hypothesized to represent a local inflammatory “cleansing” process following deep tissue B-cell depletion [51[■]]. Recognizing LICATS as a separate toxicity entity is important so as to avoid inappropriate use of immunosuppression yet should be systematically captured in future trials of CAR T and other deep B-cell-depleting therapies in autoimmune disease [51[■]].

Table 1. Summary of car T related toxicities

Toxicity	Nature	Symptoms	Management	Timing
Cytokine release syndrome (CRS)	Systemic inflammatory response	Fever, hypotension, organ dysfunction	Supportive care, tocilizumab, corticosteroids	~1–7 days post-infusion
Immune effector cell-associated neurotoxicity syndrome (ICANS)	Neurological toxicity	Headache, confusion, seizures	Supportive care, corticosteroids, anti-seizure medications	~1–7 days post-infusion ^a
Local immune effector cell-associated toxicity syndrome (LICATS)	Immune-mediated reset toxicity related to clearance	Disease specific organ inflammation	self-limited Symptomatic treatment	Days to weeks
Immune effector cell (IEC)-associated enterocolitis	Cellular inflammation of the intestines with intra-epithelial lymphocytosis and villous blunting	Diarrhea, abdominal pain, colitis	Supportive care, anti-inflammatory medications, immunosuppressive agents (e.g. TNF inhibitors integrin blockers)	Variable, often weeks post-infusion

^aNot observed in the CASTLE study.

In addition to CRS and LICATs, an immune effector cell-associated enterocolitis following chimeric antigen receptor T-cell therapy has been reported in several hematologic CAR T studies leading the FDA to issue update black box warnings for use of these agents in hematologic malignancy [52]. In a multiple myeloma CAR T study, one patient where CAR-specific immunofluorescence stains were available, CAR T-cell presence was confirmed within the lamina propria [53].

Regulators also have issued class-wide warnings about secondary T-cell malignancies following BCMA- and CD19-directed CAR-T in oncology, so emerging rheumatology series emphasize the need for long-term surveillance of autoimmune recipients [31–33,37–40,41¹¹,42–45]. Newer CAR designs incorporating non-integrating vectors, suicide switches, or transient RNA-based expression aim to reduce persistent insertional events while preserving the capacity to induce durable remission [7,8,38,43,46,47¹¹,48¹¹,49¹¹,50,54¹¹,55,56¹¹]34–36].

BEYOND SLE: IIM, SSC AND NEUROLOGIC AUTOIMMUNITY

Beyond SLE, several reports describe CD19 CAR-T as rescue therapy for severe, refractory antisynthetase syndrome and other IIMs after failure of multiple B-cell-depleting antibodies and advanced immunosuppressive regimens. In these cases, CAR-T treatment led to major improvements in muscle strength, normalization of creatine kinase, and resolution of lung involvement, sometimes with later transient flares attributed to expansion of autoreactive CD8⁺ effector cells likely driven by low-grade LICATs [51¹¹]. These observations highlight both the promise of CAR-T in IIM and the need for close T-cell monitoring and tailored inclusion criteria.

In SSC, early CD19 CAR-T experience suggests that a single infusion can stabilize or improve skin fibrosis, digital ischemia, and pulmonary function without ongoing background immunosuppression. Concurrently, cell-therapy strategies, including CAR-T and other B-cell-directed products, are being explored in SSC and fibrosing autoimmune conditions in early-phase trials [37–40,48,49¹¹].

BCMA- and CD19-targeted CAR-T strategies have also been applied to neuromyelitis optical spectrum disorder and myasthenia gravis, with phase 1 studies reporting marked reductions in pathogenic autoantibodies, relapse rates, and disability scores [34¹¹–36]. Although these indications fall outside classic rheumatology, they support the broader principle that B-lineage-directed cell therapy can reset pathogenic humoral immunity across diverse autoantibody-mediated diseases [27,28,29¹¹,30¹¹,34¹¹,35¹¹,44].

TRANSIENT AND ALTERNATIVE EFFECTOR PLATFORMS

To mitigate long-term risks and bypass procedures necessary for autologous CAR T, transient CAR-T approaches using RNA electroporation or mRNA vectors have been developed [34–36¹¹,38,43]. In a phase 1b/2a study in myasthenia gravis, ex vivo RNA-engineered BCMA CAR-T allowed patients to remain on baseline immunosuppression, eliminated the need for lymphodepleting conditioning, enabled outpatient infusion, and yielded clinically meaningful improvements with minimal CRS or ICANS. In another study an engineered CD8 T-cell-targeting LNP encapsulating CD19 CAR messenger RNA was tested in five patients with severe SLE in repeated doses. Each patient demonstrated CAR generation and B-cell depletion without high grade CRS [50].

Alternative effector cell types, including gamma-delta T-cells, CAR-natural killer (NK), and CAR-T regulatory cells (CAR-Treg), are also under active preclinical and early-clinical investigation [38–40,42,43,46,47¹¹,48¹¹,59]. CAR-NK and gamma-delta cells may combine potent cytotoxicity with a lower risk of severe CRS, ICANS, and graft-versus-host disease, while CAR-Treg aim to enforce antigen-specific immune tolerance rather than simply depleting B cells [42,43,46,47¹¹,48¹¹,59]. Although data in rheumatic indications remain limited, these platforms could ultimately broaden the therapeutic toolbox and may be particularly attractive for diseases where restoring tolerance is the primary objective.

EMERGING ALLOGENEIC AND “OFF-THE-SHELF” APPROACHES

Autologous CAR-T manufacturing is complex, time-consuming, and expensive, which constrains access to patients and scalability [7,24¹¹,25,26¹¹,31,36¹¹,37–40,41¹¹,42–44]. Allogeneic, gene-edited CAR products derived from healthy donors offer a potential solution by providing standardized potency and rapid availability. Early autoimmune experience with donor-derived CD19 CAR-T suggests feasibility and encouraging efficacy, although long-term safety and immunogenicity are still being defined. Cell types such as gamma delta T-cells and NK cells are being explored as alternative allogeneic cell types [37–40,41¹¹,42–44,47¹¹,48¹¹].

Recent reviews of the CAR-T trial landscape in autoimmune rheumatic diseases document multiple allogeneic CD19 and BCMA CAR-T trials, often run alongside autologous cohorts [37–40,41¹¹,42–44]. If durable remission can be achieved with short-lived CAR exposure, allogeneic products may prove particularly attractive, as their finite persistence reduces long-term risks while still

delivering intense, time-limited B-lineage depletion [37–40,41[■],42–44,47[■],48[■],59].

Bispecific T cell engagers (TCEs), mainly CD3 × B-cell-targeting antibodies, are emerging as a rapid, deep B-cell-depleting strategy for refractory rheumatic disease, with early human data now available. Blinatumomab use (CD19 × CD3) in small RA cohorts has produced profound B-cell depletion, immune “reset” toward naïve B cells, and rapid improvements in clinical activity and imaging, with mainly low-grade infusion-related cytokine symptoms and no high grade CRS reported. Broader programs using CD19- and BCMA-directed TCEs (for RA, systemic sclerosis, dermatomyositis/idiopathic inflammatory myopathy, and primary Sjögren) similarly suggest feasibility and a favorable short-term safety profile [54[■],55,56[■]]. Early phase SLE and lupus nephritis trials are similarly proceeding.

TRIAL DESIGN AND RHEUMATOLOGY-SPECIFIC GUIDANCE

The rapid expansion of CAR-T trials in rheumatology has prompted the development of discipline-specific guidance on how to initiate and manage such studies. A recent consensus document from the Lupus Clinical Investigators Network outlines practical recommendations for trial design in autoimmune rheumatic diseases, including patient selection, timing and conditions of leukapheresis, management of background immunosuppression and glucocorticoids, CRS and ICANS grading and management, infection prophylaxis, and long-term follow-up. Together with oncology-derived toxicity frameworks, this guidance provides a blueprint for harmonized, multi-center ARD CAR-T trials [24[■],25,26[■],31–33,37–40,41[■],42–46,51[■],52,53].

Concurrently, systematic reviews and state-of-the-art articles have begun to synthesize early clinical experience across autoimmune rheumatic diseases, placing individual case series and small cohorts in a broader context [36[■],37–40,43]. These reviews emphasize consistently high remission rates in refractory SLE, promising signals in SSc and IIM, and a generally manageable safety profile when CAR-T is delivered in experienced centers [27,28,29[■],30[■],36[■],37–40,41[■],46,49[■],58].

POSITIONING CELL THERAPIES WITHIN RHEUMATOLOGY

Current evidence supports B-lineage-directed CAR-T therapy as a powerful option for selected patients with severe, refractory autoimmune rheumatic disease who have failed treatment with available biologic and small-molecule therapies [27,28,29[■],30[■],34–36[■],37–

40,43–45,47[■],48,49[■],50]. Unlike chronic immunosuppressive regimens, CAR-T can induce profound B-cell depletion followed by reconstitution of a qualitatively different B-cell pool, with durable remission observed even after B-cell recovery, consistent with a “reset” of humoral memory [29[■],30[■],36[■],37–40,47[■],48,49[■],50].

At the same time, cost, infrastructure requirements, and long-term safety uncertainties limit near-term applicability, and equitable access will be a central ethical challenge [15–17,31,37–40,41[■],42–45]. The field is now moving from case reports and single-center series toward structured, guideline-informed clinical trials that will better define patient selection, comparative effectiveness versus optimized biologic regimens, and the role of newer platforms such as allogeneic CAR-T, CAR-NK, and TCEs [36[■],37–40,41[■],42,43,46,47[■],48[■],54[■],55,56[■]].

For rheumatology, the emergence of cell therapy represents an opportunity to harness a transformative technology capable of deep, drug-free remission in otherwise intractable disease. It will be our responsibility to ensure careful patient selection, rigorous long-term monitoring, and a focus on safety, equity, and sustainability.

Acknowledgements

None.

Financial support and sponsorship

None.

Conflicts of interest

Lupus Research Alliance, Lupus Clinical Investigators Network, Adicet Bio, Alumis Inc., ICON Labs., Cabaletta Bio, BMS, Novartis, Astra Zeneca.

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